Day 048 | Unit 04 | Teacher plan

Numeric and Algebraic Expressions

Find Greatest Common Factors

Standards: 6.NS.B.4a

Learning target: I can find the greatest common factor and interpret equal groups.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. List factor pairs or use prime factorizations. The greatest factor shared by both numbers is the GCF. For identical kits with no leftovers, the GCF gives the greatest number of kits.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 17

Day 048 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Find Greatest Common Factors

Standards: 6.NS.B.4a

Learning target: I can find the greatest common factor and interpret equal groups.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

01 / Learn

List factor pairs or use prime factorizations. The greatest factor shared by both numbers is the GCF. For identical kits with no leftovers, the GCF gives the greatest number of kits.

Worked example 1

Find the GCF of 6 and 9. Show factor pairs or prime factors.

1. List factor pairs or use prime factorizations.

2. The greatest factor shared by both numbers is the GCF.

3. Common factors: 1, 3

4. greatest is 3.

5. State the result in the form requested: 3.

Answer: 3

Worked example 2

Select all true statements about 36 and 54. Explain why the greatest factor fits the kit situation.
A. 90 divides both counts.
B. 19 is their greatest common factor.
C. 18 divides both 36 and 54.
D. 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

1. List factor pairs or use prime factorizations.

2. GCF(36, 54) = 18.

3. Divide each count by 18: 2 pencils and 3 erasers per kit.

4. A larger shared divisor is impossible.

Answer: C, D - 18 divides both 36 and 54.; 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

Day 048 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find the GCF of 8 and 12. Show factor pairs or prime factors.

02 / Constructed response

Find the GCF of 10 and 15. Show factor pairs or prime factors.

03 / Constructed response

Find the GCF of 12 and 16. Show factor pairs or prime factors.

04 / Constructed response

Find the GCF of 45 and 60. Show factor pairs or prime factors.

05 / Constructed response

Find the GCF of 70 and 100. Show factor pairs or prime factors.

06 / Constructed response

Find the GCF of 28 and 42. Show factor pairs or prime factors.

07 / Constructed response

Make the greatest possible number of identical kits from 24 pencils and 36 erasers with no leftovers. Find the number of kits and each kit’s contents.

08 / Multiple select

Select all true statements about 18 and 27. Explain why the greatest factor fits the kit situation.
A. 9 is the largest number of identical no-leftover kits from 18 pencils and 27 erasers.
B. 10 is their greatest common factor.
C. 9 divides both 18 and 27.
D. 45 divides both counts.

09 / Multiple select

Select all true statements about 30 and 45. Explain why the greatest factor fits the kit situation.
A. 15 is the largest number of identical no-leftover kits from 30 pencils and 45 erasers.
B. 15 divides both 30 and 45.
C. 16 is their greatest common factor.
D. 75 divides both counts.

10 / Constructed response

Make the greatest possible number of identical kits from 18 pencils and 30 erasers with no leftovers. Find the number of kits and each kit’s contents.

Day 048 | Unit 04 | Answer key

Numeric and Algebraic Expressions

Find Greatest Common Factors

Standards: 6.NS.B.4a

Learning target: I can find the greatest common factor and interpret equal groups.

1. 4

Common factors: 1, 2, 4; greatest is 4.

6.NS.B.4a | Constructed response

2. 5

Common factors: 1, 5; greatest is 5.

6.NS.B.4a | Constructed response

3. 4

Common factors: 1, 2, 4; greatest is 4.

6.NS.B.4a | Constructed response

4. 15

Common factors: 1, 3, 5, 15; greatest is 15.

6.NS.B.4a | Constructed response

5. 10

Common factors: 1, 2, 5, 10; greatest is 10.

6.NS.B.4a | Constructed response

6. 14

Common factors: 1, 2, 7, 14; greatest is 14.

6.NS.B.4a | Constructed response

7. 12 kits; 2 pencils and 3 erasers each

GCF(24, 36) = 12; divide both counts by 12.

6.NS.B.4a | Constructed response

8. A, C - 9 is the largest number of identical no-leftover kits from 18 pencils and 27 erasers.; 9 divides both 18 and 27.

GCF(18, 27) = 9. Divide each count by 9: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

9. A, B - 15 is the largest number of identical no-leftover kits from 30 pencils and 45 erasers.; 15 divides both 30 and 45.

GCF(30, 45) = 15. Divide each count by 15: 2 pencils and 3 erasers per kit. A larger shared divisor is impossible.

6.NS.B.4a | Multiple select

10. 6 kits; 3 pencils and 5 erasers each

GCF(18, 30) = 6; divide both counts by 6.

6.NS.B.4a | Constructed response